

Module 1 Lesson 3

Lesson Tasks

Question:

You have recently acquired a new business and your business requires the services of a courier / delivery company to be able to deliver your goods in a timely manner during different weather conditions. The pay off table states what the delivery company will be charging your business depending on their facilities to be able to accommodate certain weather conditions.

The pay-off table for this dilemma has been presented in Figure 1 and all question below will relate to this illustration.

	Weather Conditions		
	Good	Average	Poor
UberFreight	19 000	9 000	5 000
CourierIT	7 000	15 000	8 000
Deliver-MAN	4 000	6 000	11 000

Figure 1: Delivery Service Pay off Table

Help the business to make the decision using the following decision criteria (Note: Remember to provide insights on your answers):

Maximax criterion

Solution:

	Weather Conditions			Best
	Good	Average	Poor	
UberFreight	19 000	9 000	5 000	19 000
CourierIT	7 000	15 000	8 000	15 000
Deliver-MAN	4 000	6 000	11 000	11 000
Maximax Criterion recommends:			UberFreight	

Wald's criterion

Solution:

	Weather Conditions			
	Good	Average	Poor	Worst
UberFreight	19000	9000	5000	5000
CourierIT	7000	15000	8000	7000
Deliver-MAN	4000	6000	11000	4000
Wald's Maximin Criterion recommends: CourierIT				

Savage's criterion

Solution:

	Weather Conditions			
	Good	Average	Poor	
UberFreight	19 000	9 000	5 000	
CourierIT	7 000	15 000	8 000	
Deliver-MAN	4 000	6 000	11 000	

	Weather Conditions			Regret
	Good	Average	Poor	
UberFreight	0	6 000	6 000	6 000
CourierIT	12 000	0	3 000	12 000
Deliver-MAN	15 000	9 000	0	15 000
Savage's Regret Minimax Criterion recommends:				UberFreight

Laplace's criterion

Solution:

	Weather Conditions			
	Good	Average	Poor	Average
UberFreight	19 000	9 000	5 000	11 000
CourierIT	7 000	15 000	8 000	10 000
Deliver-MAN	4 000	6 000	11 000	7 000
Laplaces' Criterion recommends: UberFreight				

Hurwicz's criterion with α equal to 0%, 25%, 50%, 75% and 100%

Solution:

	State of the economy			MaxMin	alpha				MaxMax
	Good	Average	Poor	0%	25%	50%	75%	100%	
UberFreight	19 000	9 000	5 000	5000	8500	12000	15500	19000	
CourierIT	7 000	15 000	8 000	7000	9000	11000	13000	15000	
Deliver-MAN	4 000	6 000	11 000	4000	5750	7500	9250	11000	
Hurwics criterion recommends:					CourierIT				

Expected payoff criterion with probabilities 20%, 50% and 30%

Solution:

State of the economy				
	Good	Average	Poor	Expected payoff
UberFreight	19 000	9 000	5 000	9800
CourierIT	7 000	15 000	8 000	11300
Deliver-MAN	4 000	6 000	11 000	7100
Probability	0.20	0.50	0.30	

Expected payoff criterion recommends: CourierIT

Expected regret with probabilities 20%, 50% and 30%

Solution:

							Weather Conditions		
		Weather Conditions				Good	Average	Poor	Expected Regret
		Good	Average	Poor	UberFreight	0	6 000	6000	4800
	UberFreight	19 000	9 000	5 000	CourierIT	12000	0	3000	3300
	CourierIT	7 000	15 000	8 000	Deliver-MAN	15000	9 000	0	7500
	Deliver-MAN	4 000	6 000	11 000	Probability	0.2	0.5	0.3	
Expected loss of opportunity recommends:				CourierIT					